

PHY102 Electricity and Magnetism Jan-April 2026: Assignment 6

Arvind, IISER Mohali, arvind@iisermohali.ac.in

To be discussed in the Tutorial on March 20, 2026

1. Consider an infinite conducting hollow cylinder of inner radius R_1 and outer radius R_2 . Imagine a thin wire with linear charge density λ passing through the center of the cylinder. Calculate the electric field in all space. What will be the induced surface charge density on the inside and outside surface of the cylinder?
2. Consider two concentric hollow spheres of radii R_1 and R_2 , ($R_2 > R_1$). Calculate the capacity of this system. You can imagine that the inner sphere has charge $-Q$ and outer sphere has charge $+Q$ uniformly distributed over them.
3. Consider two charges $\pm q$ located at $z = \pm d/2$ i.e. distance d apart along the z direction. They form an electric dipole. By calculating the field at a few

points of your choice draw the field lines of this dipole. Try and obtain the expression of the field in terms of the dipole moment for distances larger compared to d . What is the electric field for distances very large compared to d ?

4. The force on a charged particle when both electric and magnetic fields are present is given by the Lorentz force formula

$$\vec{F} = q\vec{E} + \frac{q}{c}(\vec{v} \times \vec{B})$$

Where $\vec{v}$ is the velocity of the charged particle. Consider a particle with charge q at rest at the origin. Consider $\vec{E} = E_0\hat{x}$ and $\vec{B} = B_0\hat{z}$. Calculate the trajectory of the particle. Plot the trajectory. Calculate the kinetic energy of the particle as a function of time.